Smart Agriculture System

CROP RECOMMENDATION USING AI

(Weather+Soil Data)

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Problem **Statement**

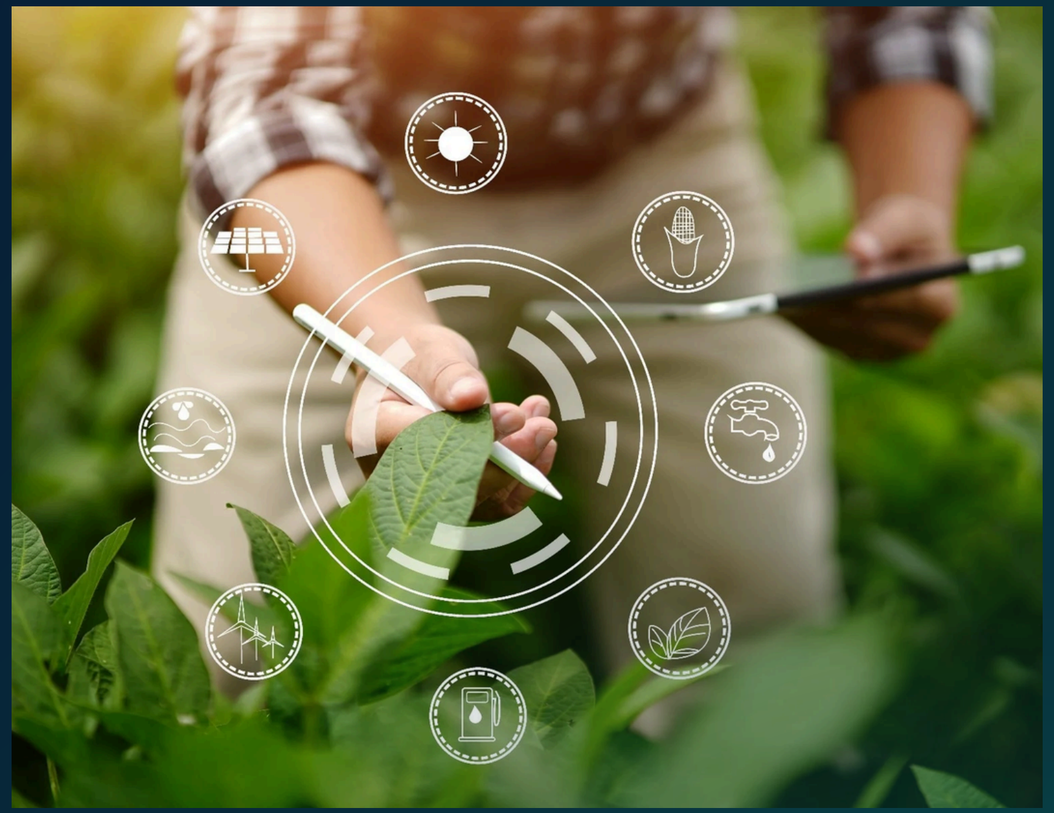
Selecting the right crop is more complex than it should be.

Agriculture depends heavily on soil nutrients and environmental conditions.
However, many farmers still rely on traditional knowledge rather than scientific data.
Lack of awareness about soil composition and unpredictable weather leads to poor crop selection, reduced yield, and financial loss.

- Improper crop selection
- Lack of soil awareness
- Unpredictable weather
- Reduced agricultural yield

Our **Solution**

An AI-based system that analyzes soil nutrients and environmental conditions to predict the best crop for farming.



One smart system for data-driven farming decisions

Our system uses soil and weather data to recommend the most suitable crop, reducing guesswork and improving productivity.

- AI-based crop prediction
- Data-driven decisions
- Easy to use

Objectives

Develop an AI-based system to recommend suitable crops using machine learning techniques

Utilize weather data (temperature, humidity, rainfall) for accurate crop prediction

Analyze soil nutrients (N, P, K, pH) to understand soil condition and support better crop selection

Improve agricultural productivity by enabling data-driven farming decisions

Societal Benefit

- Helps farmers make informed decisions
- Reduces risk of crop failure
- Increases agricultural productivity
- Promotes sustainable farming practices
- Supports efficient use of soil and water resources

Possible Solutions

Use Machine Learning (Random Forest) to train a model using agricultural dataset and predict suitable crops

Analyze soil nutrients (N, P, K, pH) and weather conditions (temperature, humidity, rainfall) as input features

Train the model on historical data to learn patterns between environmental conditions and crops

Generate accurate and fast predictions based on the trained model